# Alireza Ghorbani

*Curriculum Vitae*

+98(910)6768616
alcstat@outlook.com
alcstat.github.io
alcstat
alcstat
R&D Department, Zarrin Roya Co, Tehran 1513754415, Iran


*October 10, 2023*

## Education

2014–2016 **M.Sc. in Mathematical Statistics**, Shahid Beheshti University
- *Thesis: Analysis of High Dimensional Data Using Rayleigh Quotient Optimization (in Persian) under supervision of Prof. Faridrohani & prof. Hamedani, Grade: A (19/20)*
- *GPA: 16.10/20 (3.38/4)*

2009–2014 **B.Sc. in Statistics**, Shahid Chamran University
- *Thesis: Statistical Study for The Factors Contributing to The Academic Failure among A Group of Undergraduate Students (in Persian) under supervision of Prof. Zadkarami*
- *GPA: 13.95/20 (2.57/4) – Last Semester GPA: 16.83/20 (4/4)*

2006–2009 **High School Diploma in Mathematics and Physics**, *Shahid Beheshti High School*, Birjand, Iran
Affiliated with the National Organization for Development of Talented Students (NODET)
- *CGPA: 17.45/20 (3.83/4)*

## Publication

2023+ **Optimizing Water Pipeline Route Selection Using a Sparse Deep Neural Network and the Fuzzy VIKOR Method (case study: water pipeline transmission)**, *Ghorbani, A. Khajehzadeh, M. Seifi, F. Al-Ansari, N and Shahabi, H.* , Preprint

2023+ **Longitudinal Supply Chain Monitoring using Generalized estimating equation and Joint Optimization method (A case study in the Personal Care Industry)**, *Khajehzadeh, M. Pazhuheian, Ghorbani, A. F. Seifi, F. Madrakid, G.*, Under-Review

## Professional Experience

2021 - Now **Market Research Analyst & Statistician**, *R&D Department*, *Zarrin Roya*, Tehran, Iran

2022 – 2023 **Statistics & Machine learning Instructor**, *Learnix.ir*, Tehran, Iran

## Honors and Awards

2014–2016 **Awarded Fellowship**, *Graduate Program*, Statistics Department, Shahid Beheshti University, Tehran, Iran

2014 **Ranked 9th**, *in 19th National Scientific Olympiad (Statistics)*

2014 **Ranked 19th**, *in National Entrance Exam for Masters of Science in Actuarial Statistics, Social & Economic Statistics and Theoretical Statistics*

2009–2014 **Awarded Fellowship**, *Undergraduate Program*, Statistics Department, Shahid Chamran University, Ahwaz, Iran

2003 & 2005 Selected for study in schools of National Organization for Development of Exceptional Talents (NODET) through an exam with less than 1% acceptance rate.

## Technical skills

R Programming, Python , Matlab, SQL, LaTeX, Open BUGS

## Research Interest

- Natural Language Processing
- High-Dimensional Statistics
- Statistical Learning
- Bayesian Statistics

## Research Experience & Selected Projects

**Comparison of Sentiment Analysis Methods for Farsi Text**, Developing models (Logistic lasso Regression, Naive Bayes, Vector Space Model and LSTM) for sentiment analysis and text classification of Farsi text in , and comparing their performances. The best method selected for tracking customers' behavior over social media.

**Recommender System Suggesting Proper Product Size**, Developing a recommending system to help mothers find proper diaper size in . Selecting proper diaper size could be a challenge for young mothers that could lead to a significant amount of dissatisfaction.

**Water Pipeline Optimization using Multi-Task Sparse Neural Network Integrated with MCDM**, The costly intervention of five groups of experts that their idea regarding water pipeline was necessary for evaluation and determination of the best water pipeline was replaced by a multi-task neural network model. Finally, some newly proposed pipeline alternatives were qualified by the model and ranked by FVIKOR (developed in ).

**Sentiment Analysis Using Multiple Methods Based on Bag of Words Approach for Persian Text**, *In this project, I classified Persian text by the polarity (positive or negative opinions) expressed within them using 3 different method in* .

**High Dimensional Classification using Rayleigh Quotient**, *I have studied classification methodologies and the challenges for their implementation facing high-dimensional data. Then, I have provided numerical results (by simulation and by an actual breast cancer dataset) comparing these methods. This project was a part of my master's degree thesis and was done in and* .

**ROAD**, *The code for the paper "A ROAD to Classification in High Dimensional Space" translated from to* .

**Academic Failure Detection Model**, I have investigated the relevant factors that can lead to academic failure in order to detect vulnerable students providing them the help and supervision. This project was a part of my bachelor's degree thesis.

## Extracurricular Activities & Professional Certificates

- TensorFlow Developer Professional Certificate
  deeplearning.ai
- Natural Language Processing Specialization
  deeplearning.ai
- Bayesian Statistics Specialization
  University of California, Santa Cruz
- Statistics with Python Specialization
  University of Michigan
- PostgreSQL for Everyone Specialization
  University of Michigan
- Deep Learning Specialization
  deeplearning.ai
- Machine Learning Specialization
  Stanford University & deeplearning.ai
- Fundamentals of Reinforcement Learning
  University of Alberta
- Python 3 programming Specialization
  University of Michigan

## Test Scores

July 2020 **Scored 103 in TOEFL iBT Exam**

Reading (25/30), Listening (29/30), Speaking (25/30), Writing (24/30)

## Languages

- English: Fluent
- Spanish: Elementary
- Persian: Native